



BENG 280A/ECE 207A Project 2: Partial Fourier Reconstruction

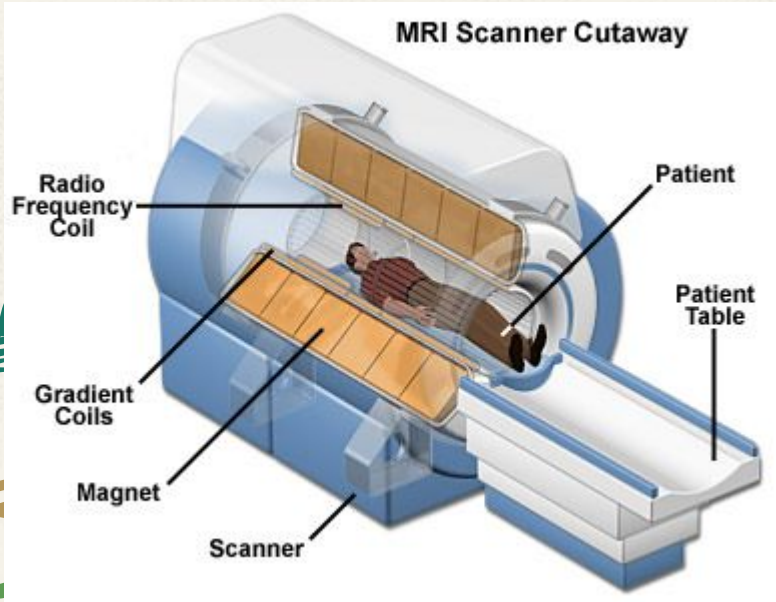
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Introduction and Background

Magnetic Resonance Imaging (MRI)



- Strong magnetic field
- Radiofrequency (RF) pulse excitation
- Gradient coil spatial encoding
- Integrate gradient directions to create MRI image [2]

Figure 1. MRI scanner and components [1]

k-Space

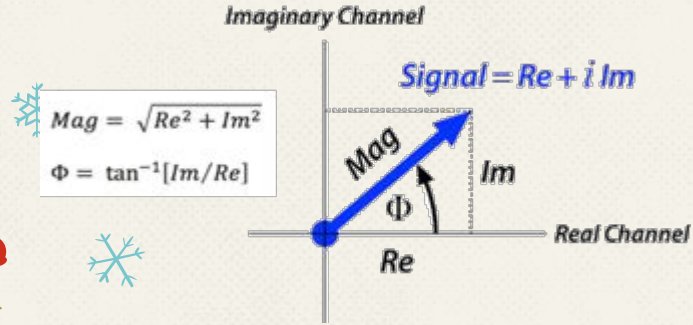


Figure 2. Vector form of frequency and phase [3]

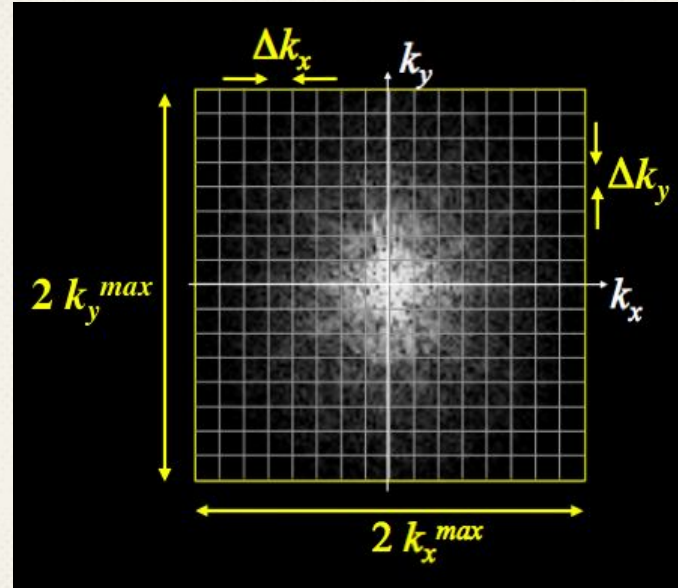
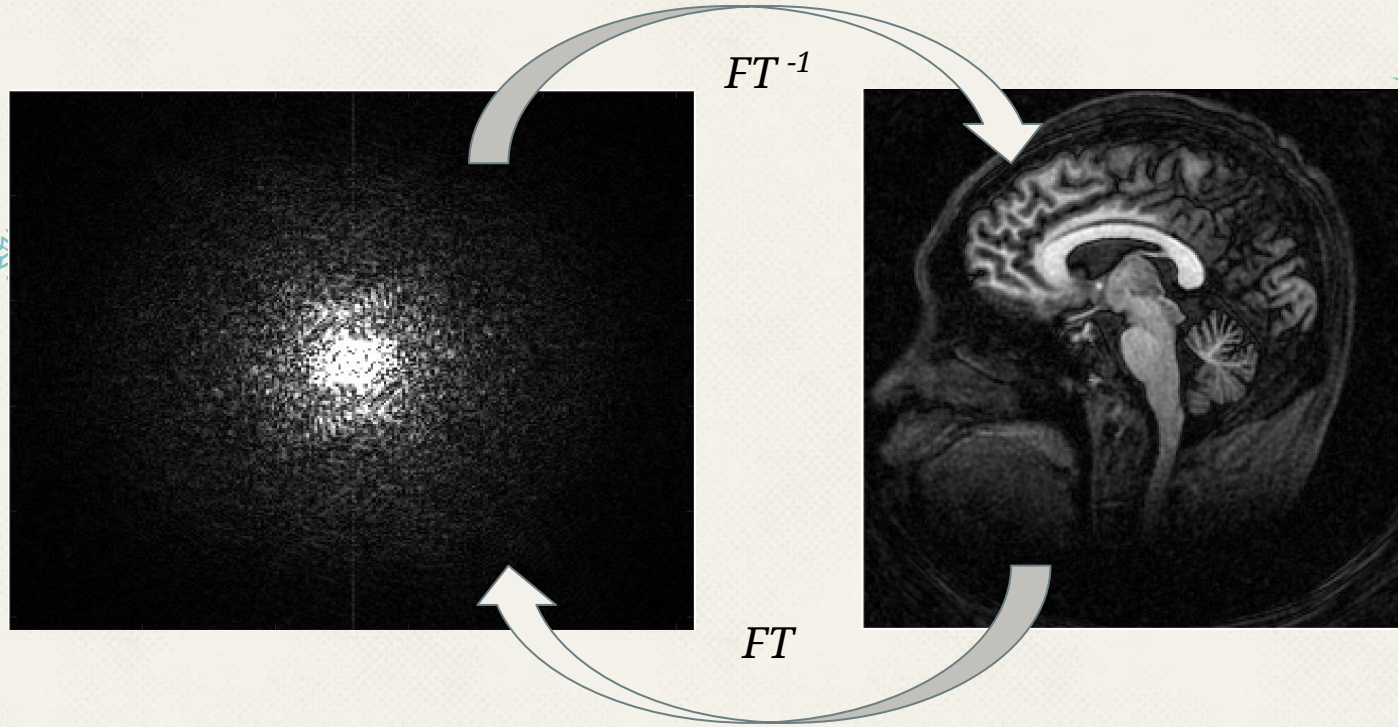


Figure 3. Depiction of *k*-space of an object [4]

Inverse Fourier Image Reconstruction



Conjugate Symmetry

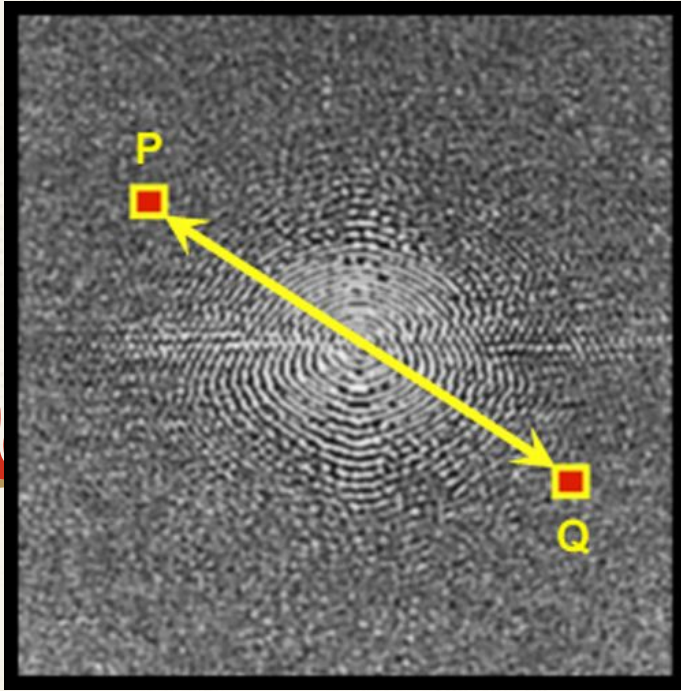


Figure 4. Conjugate symmetry of k -space [5]

- The k -space of a real image is conjugate symmetric

$$P = (-k_x, k_y) = -a+bi$$

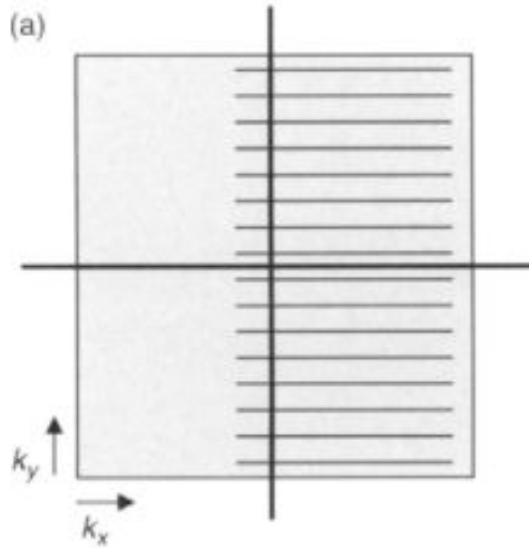
$$Q = (k_x, -k_y) = a-bi$$

$$P = -Q^*$$

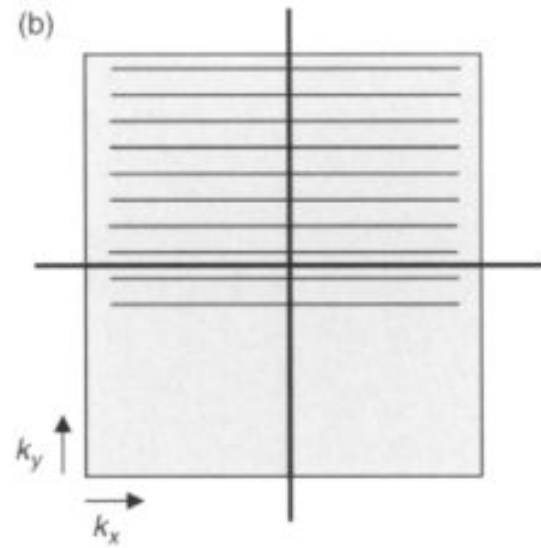
$$f(-x) = f^*(x) = \textit{Hermitian}$$

Partial Fourier Reconstruction

(PEP)



Frequency-Encoded
Direction (Partial Echo)

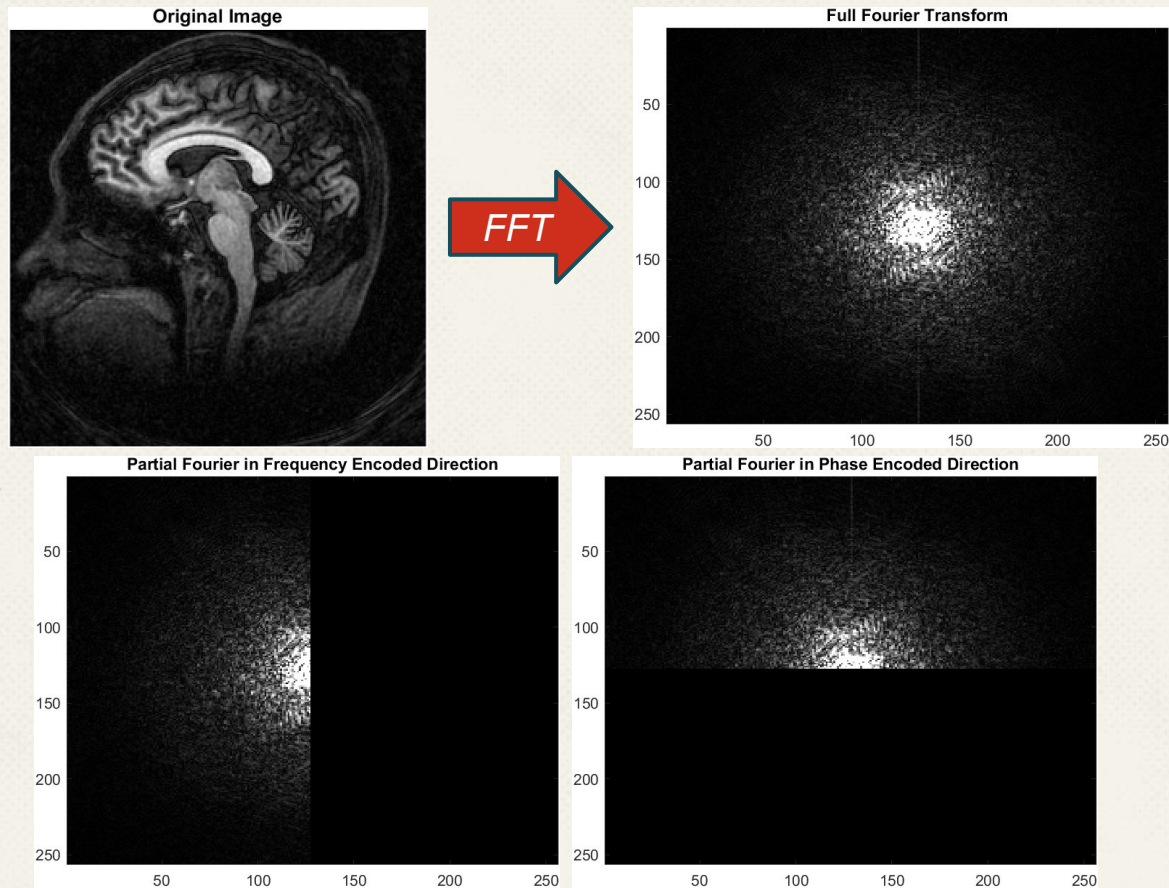


Phase-Encoded
Direction (Partial NEX) [6]



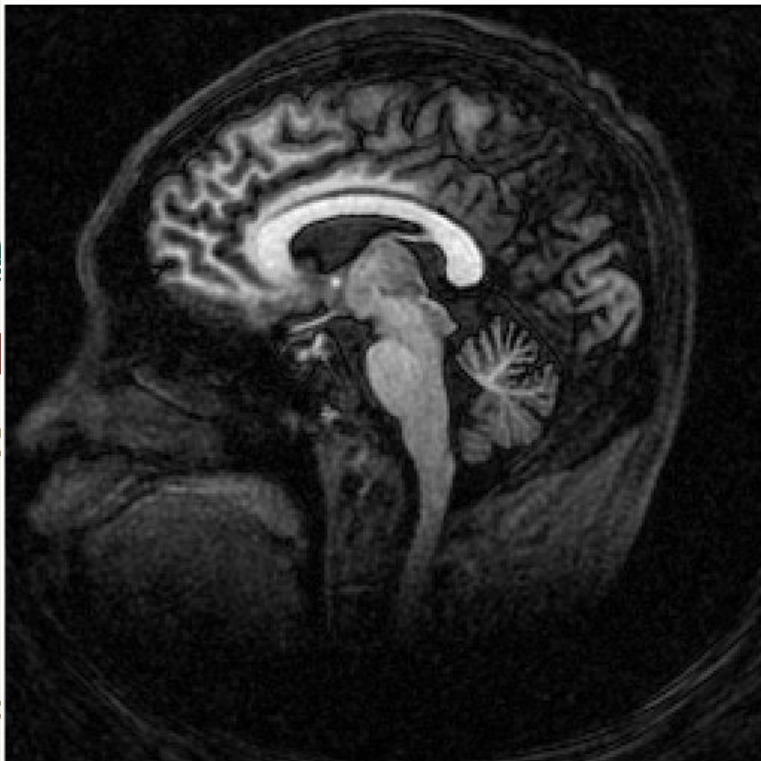
Zero-Filling

Zero-Filling Process

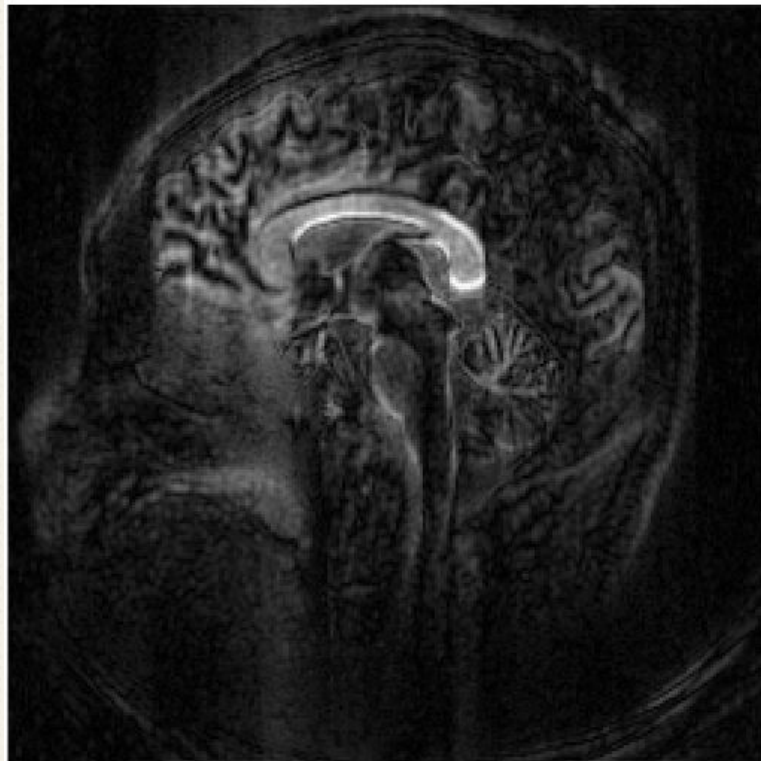


0.5 Fraction PFR

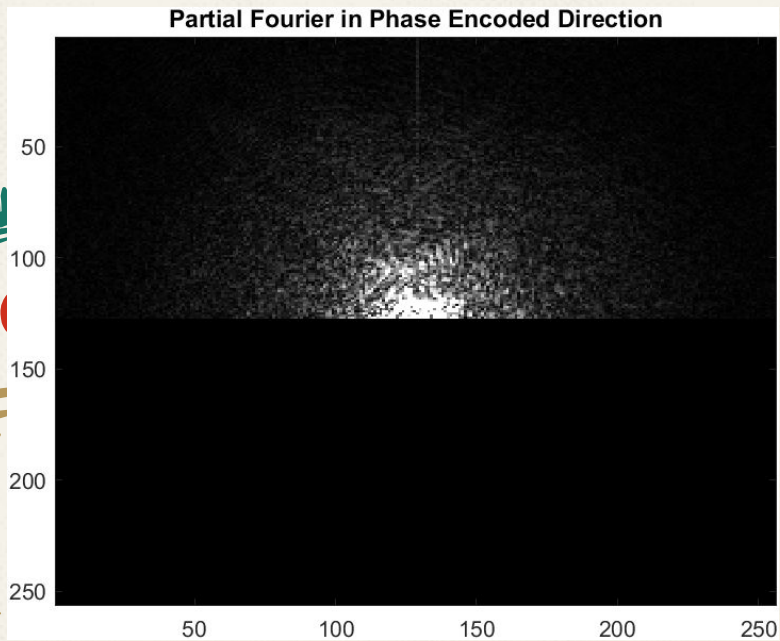
Original Image



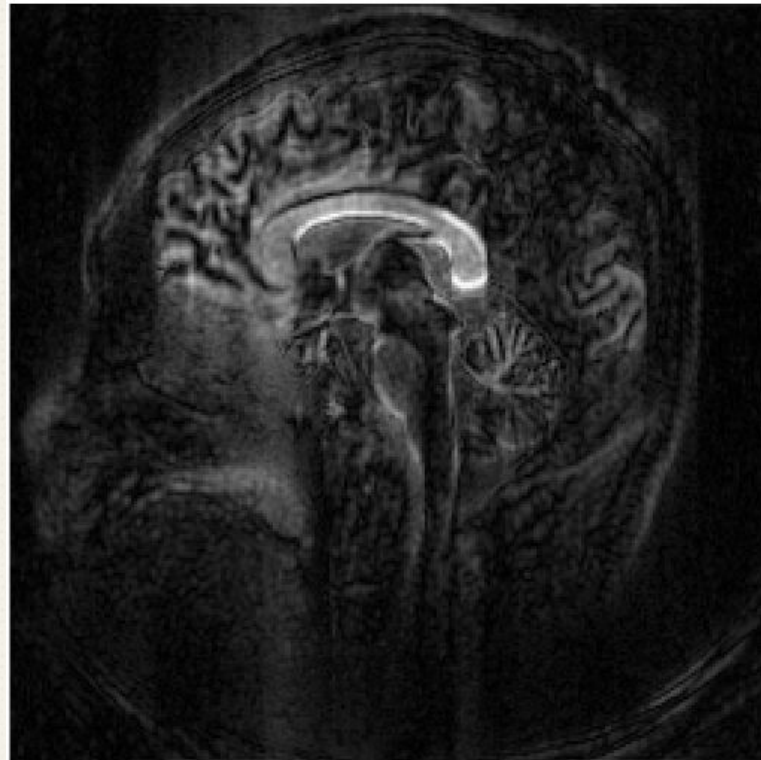
Reconstructed Image at 0.5 Without Hanning



0.5 Fraction PFR

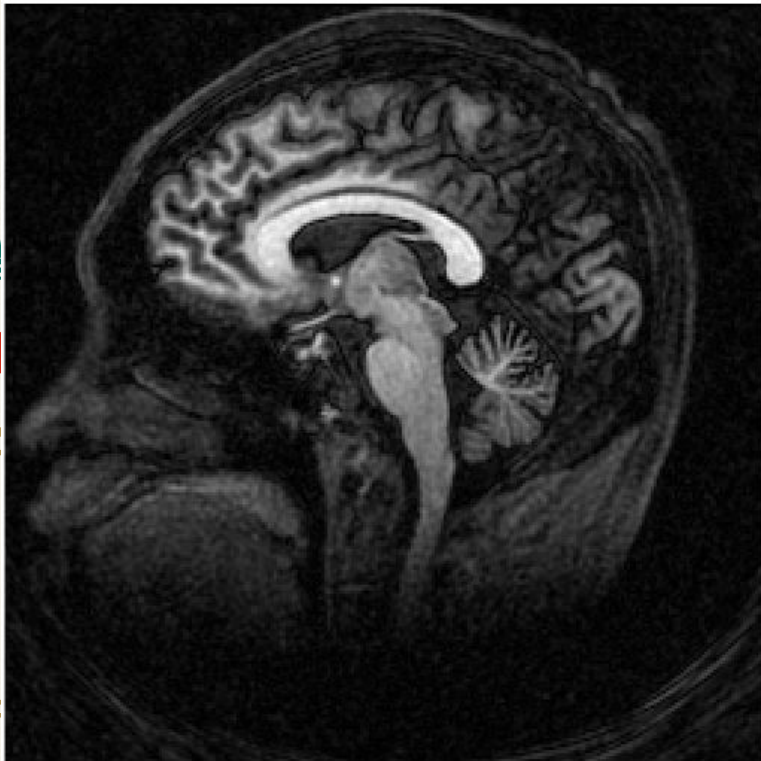


Reconstructed Image at 0.5 Without Hanning

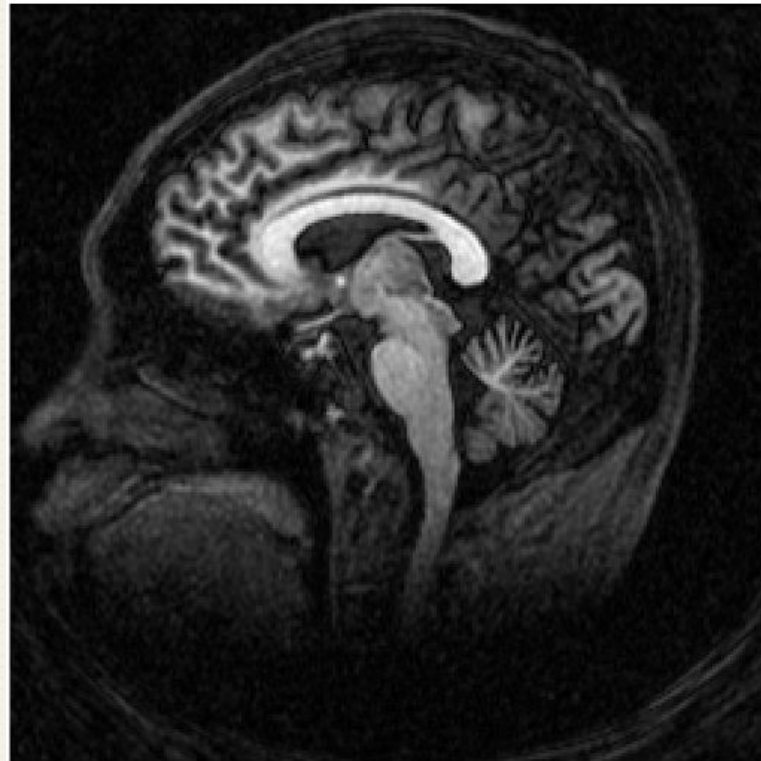


0.75 Fraction PFR

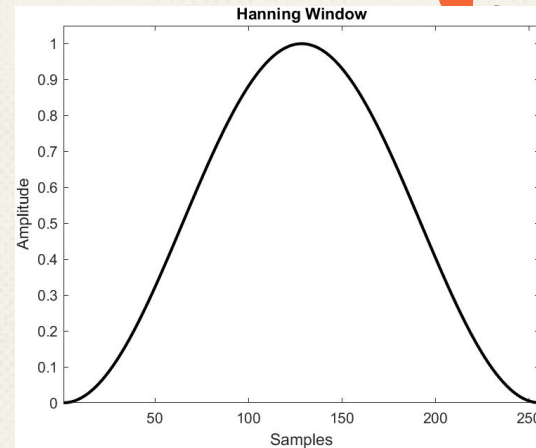
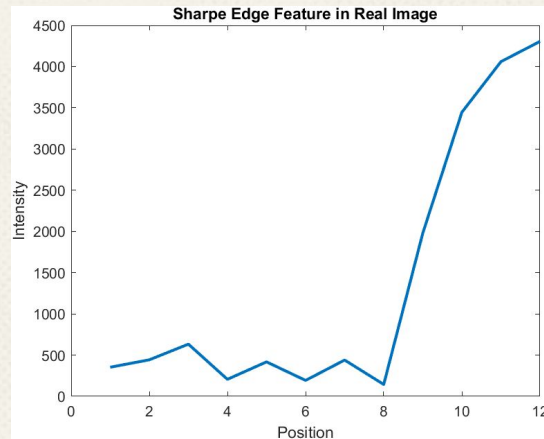
Original Image



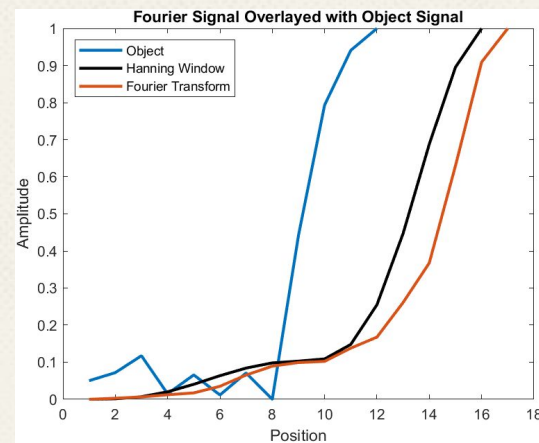
Reconstructed Image at 0.75 Without Hanning



Hanning Window



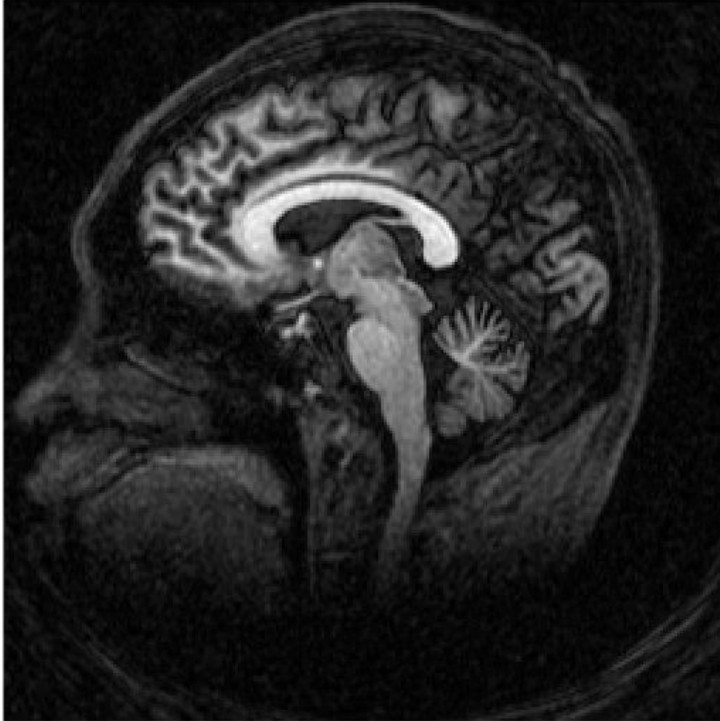
*



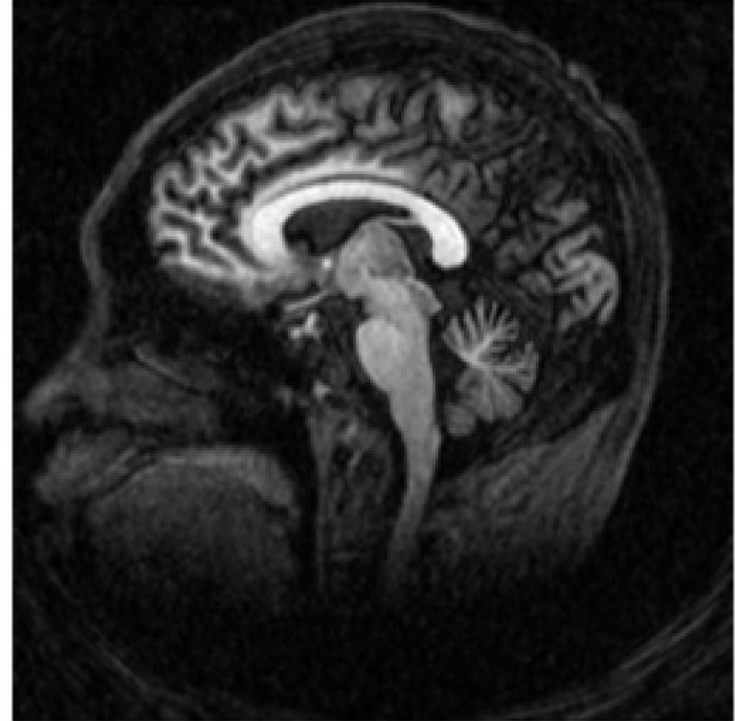
0.75 Fraction PFR - Hanning



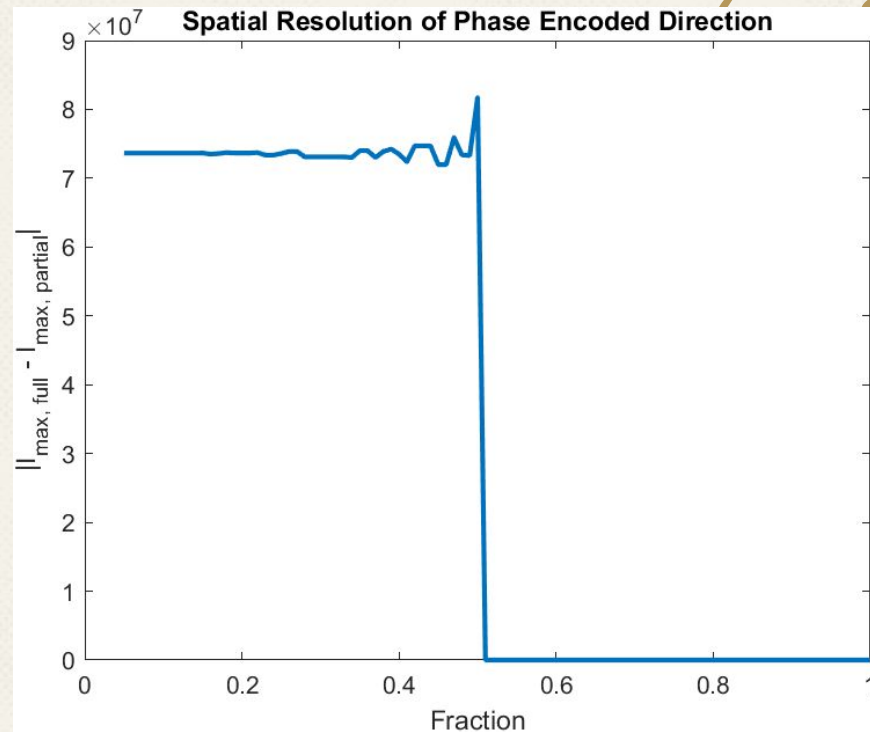
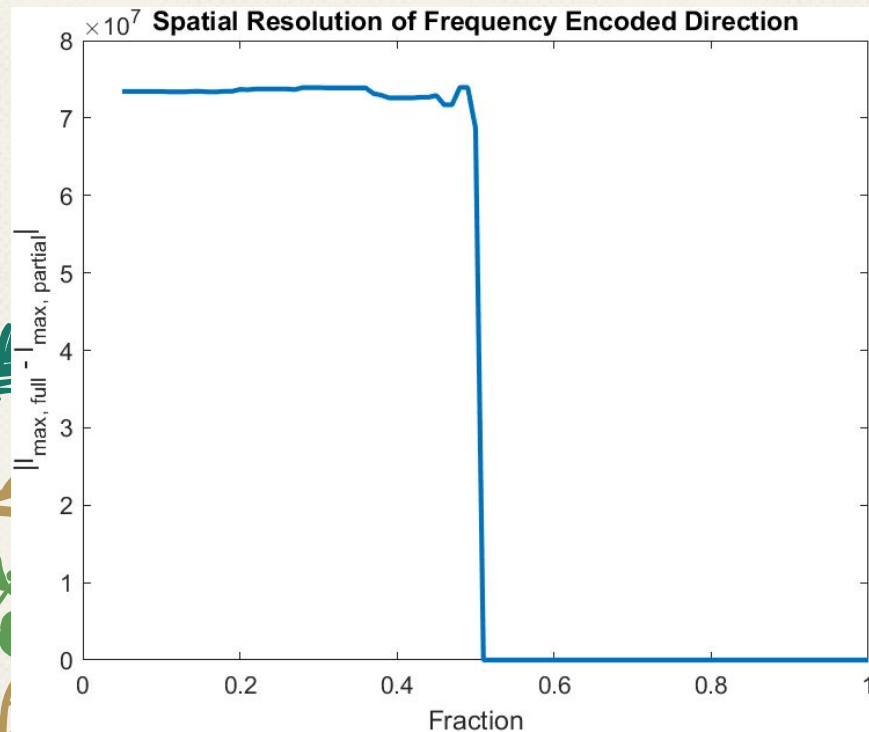
Reconstructed Image at 0.75 With Hanning (Phase)



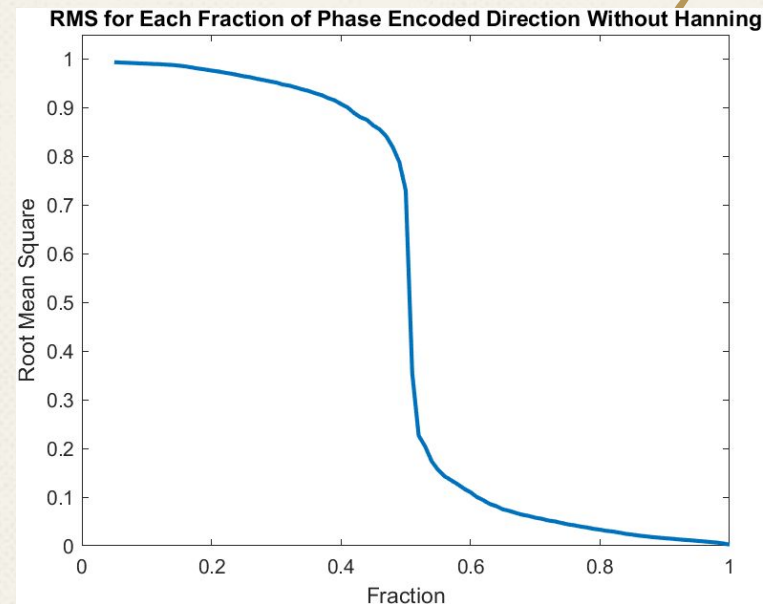
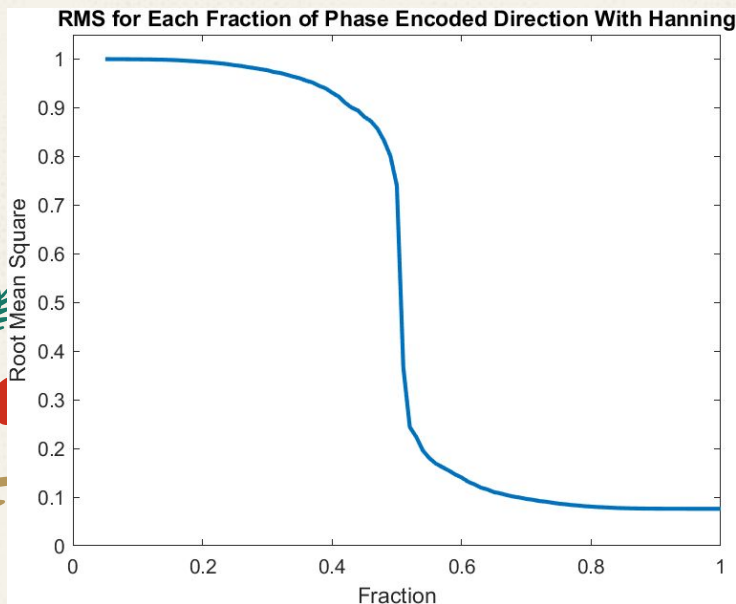
Reconstructed Image at 0.75 With Hanning (Frequency)



Spatial Resolution Difference Per Fraction



Root Mean Square Error Comparison



$$RMSE = \sqrt{\frac{1}{n^2} \sum_i^n \sum_j^n ((x_i, y_j)_{\text{partial}} - (x_i, y_j)_{\text{original}})^2}$$



Conjugate Synthesis

Conjugate Synthesis Reconstruction

Conjugate Symmetry

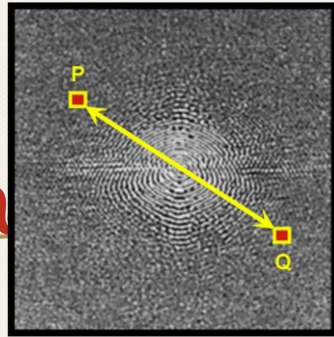


Figure 4. Conjugate symmetry of k-space [5]

- The k -space of a real image is conjugate symmetric

$$P = (-k_x, k_y) = -a + bi$$

$$Q = (k_x, -k_y) = a - bi$$

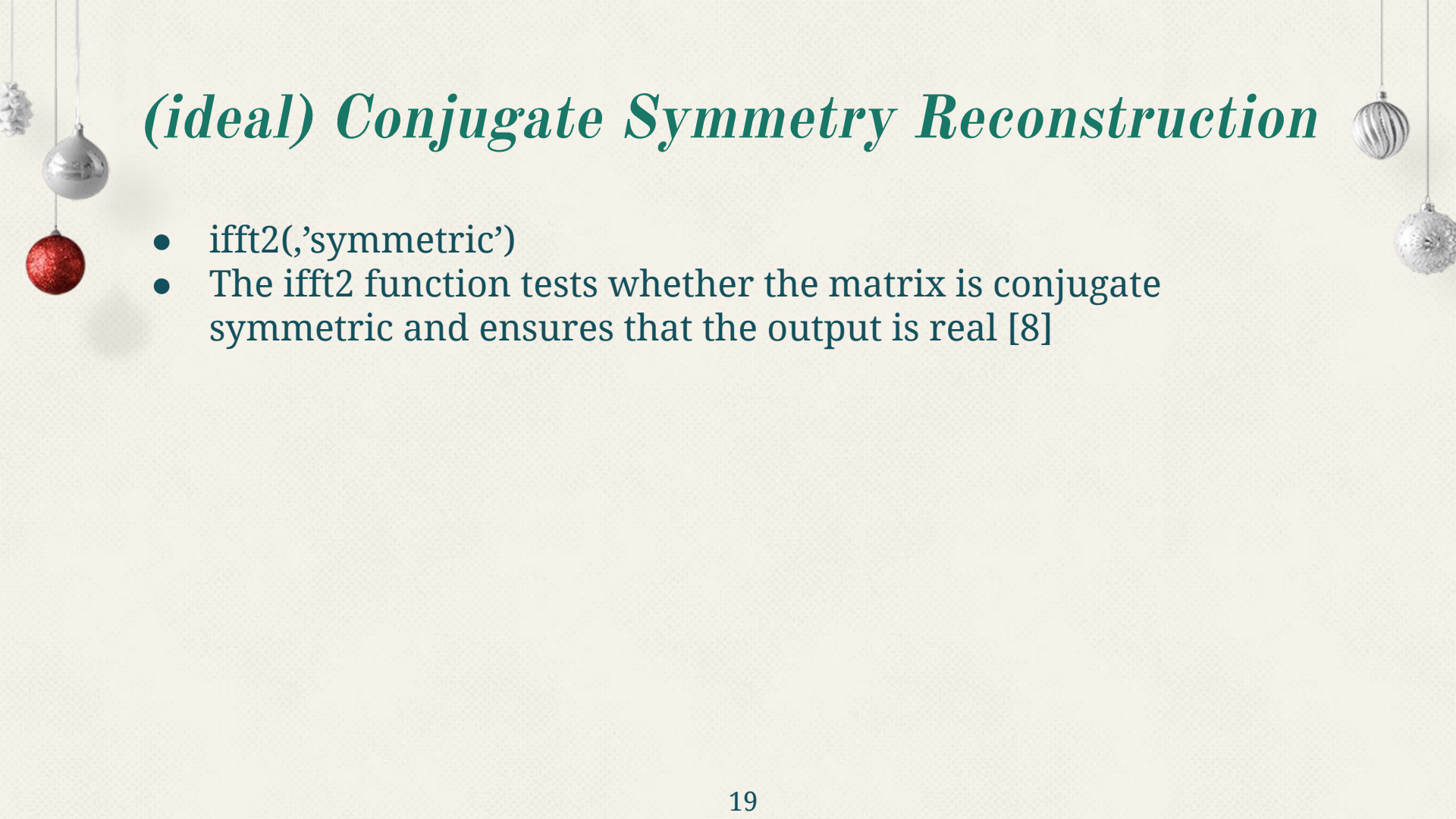
$$P = -Q^*$$

$$f(-x) = f^*(x) = \text{Hermitian}$$

- Coils under normal imaging situations are complex

Therefore

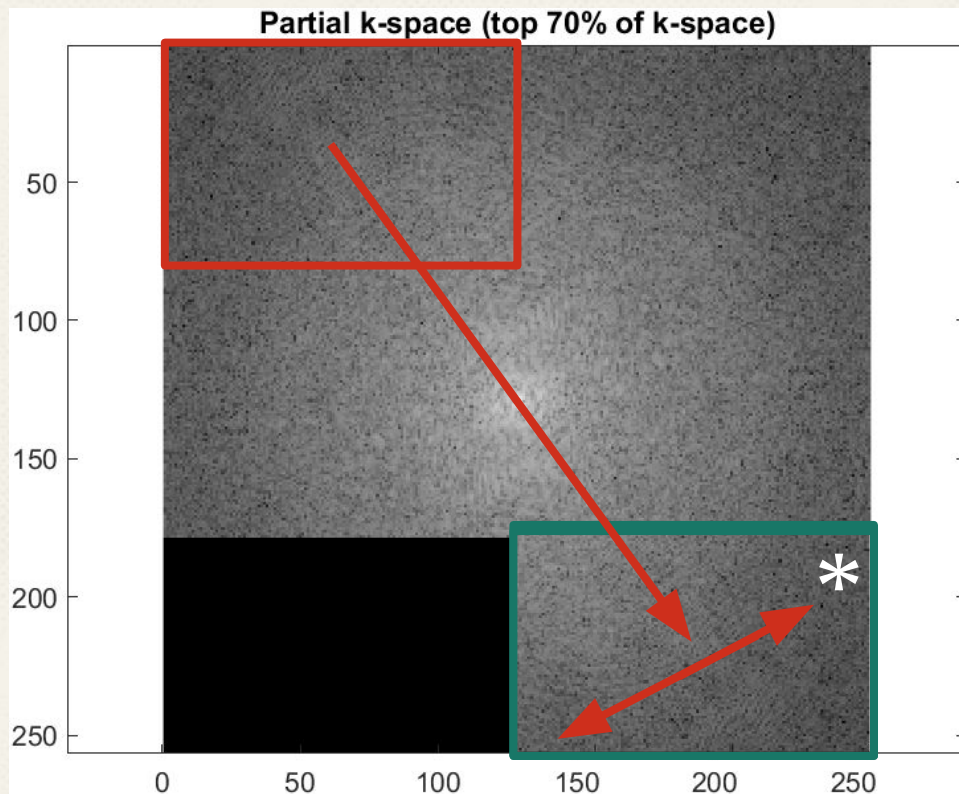
$$f(-x) \neq f^*(x) \text{ Blaimer [7]}$$



(ideal) Conjugate Symmetry Reconstruction

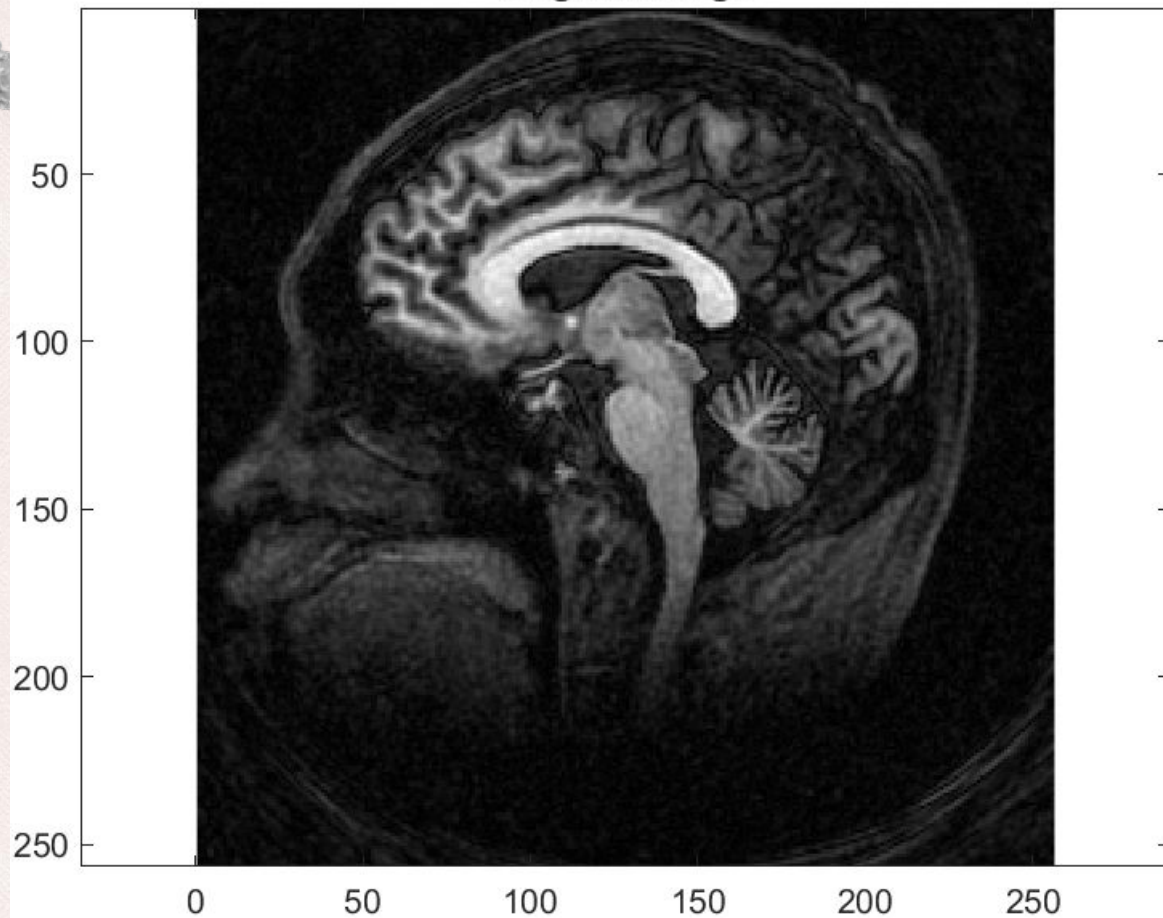
- `ifft2('symmetric')`
- The `ifft2` function tests whether the matrix is conjugate symmetric and ensures that the output is real [8]

Conjugate Synthesis Method

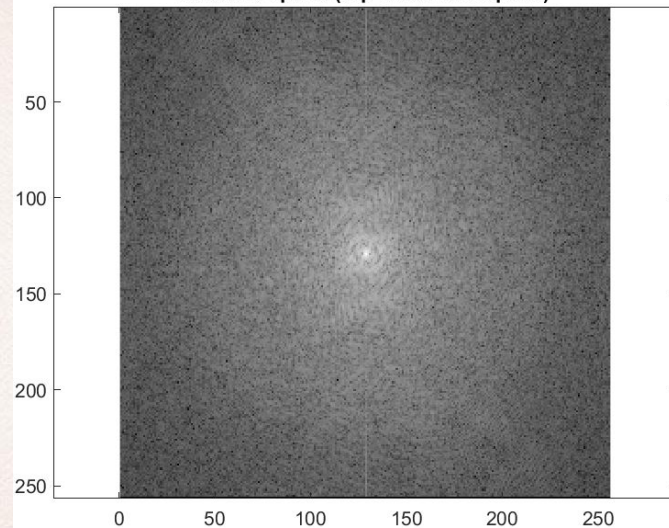


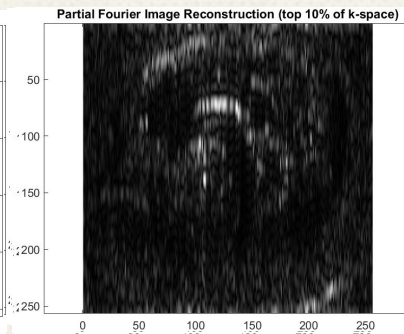
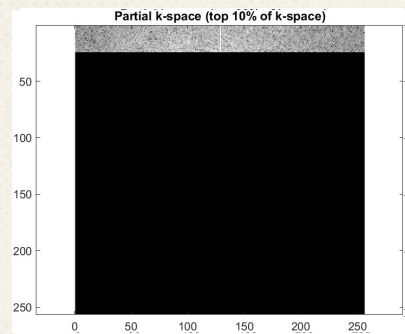
[9]

Original Image

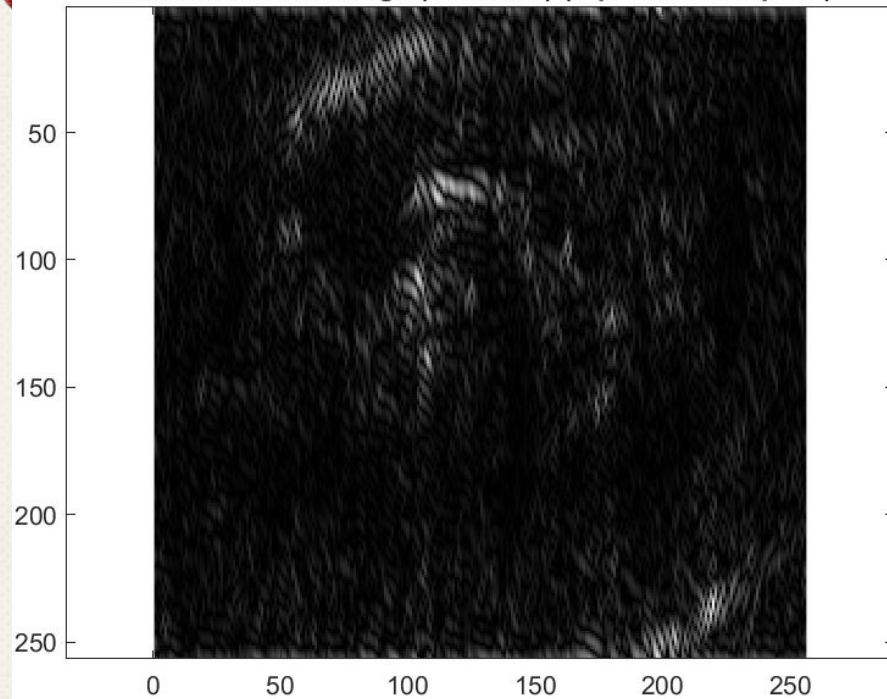


Partial k-space (top 100% of k-space)

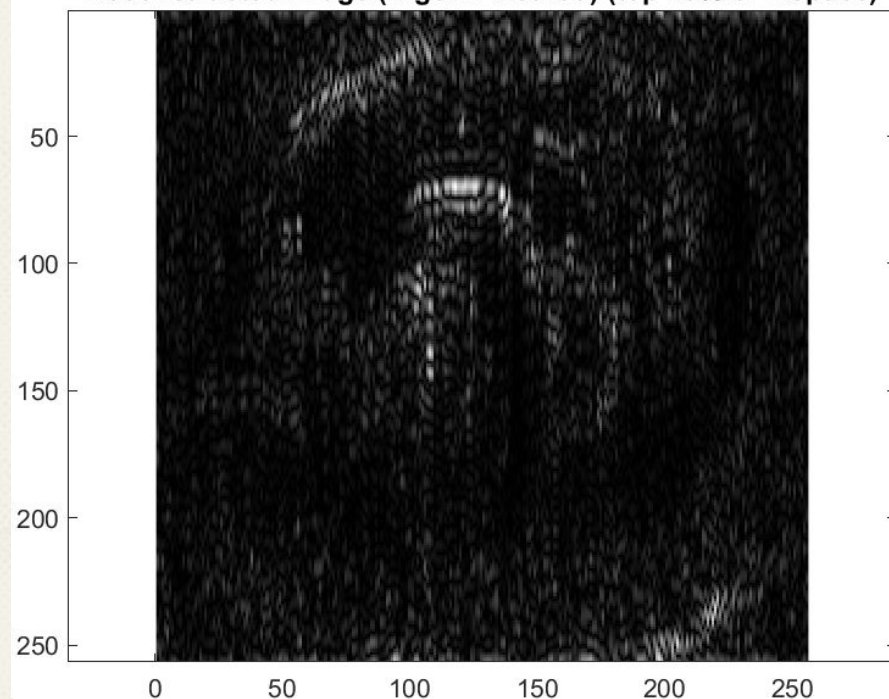


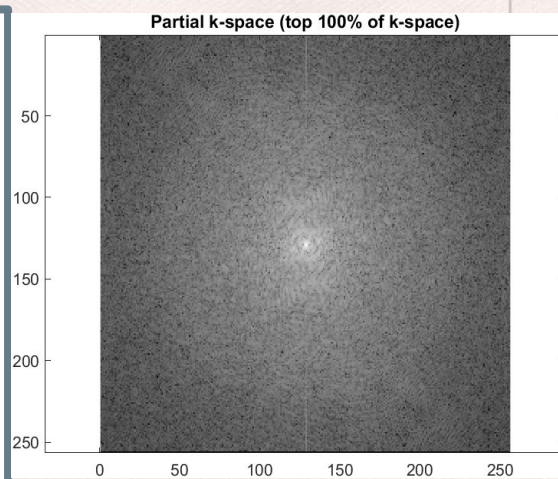
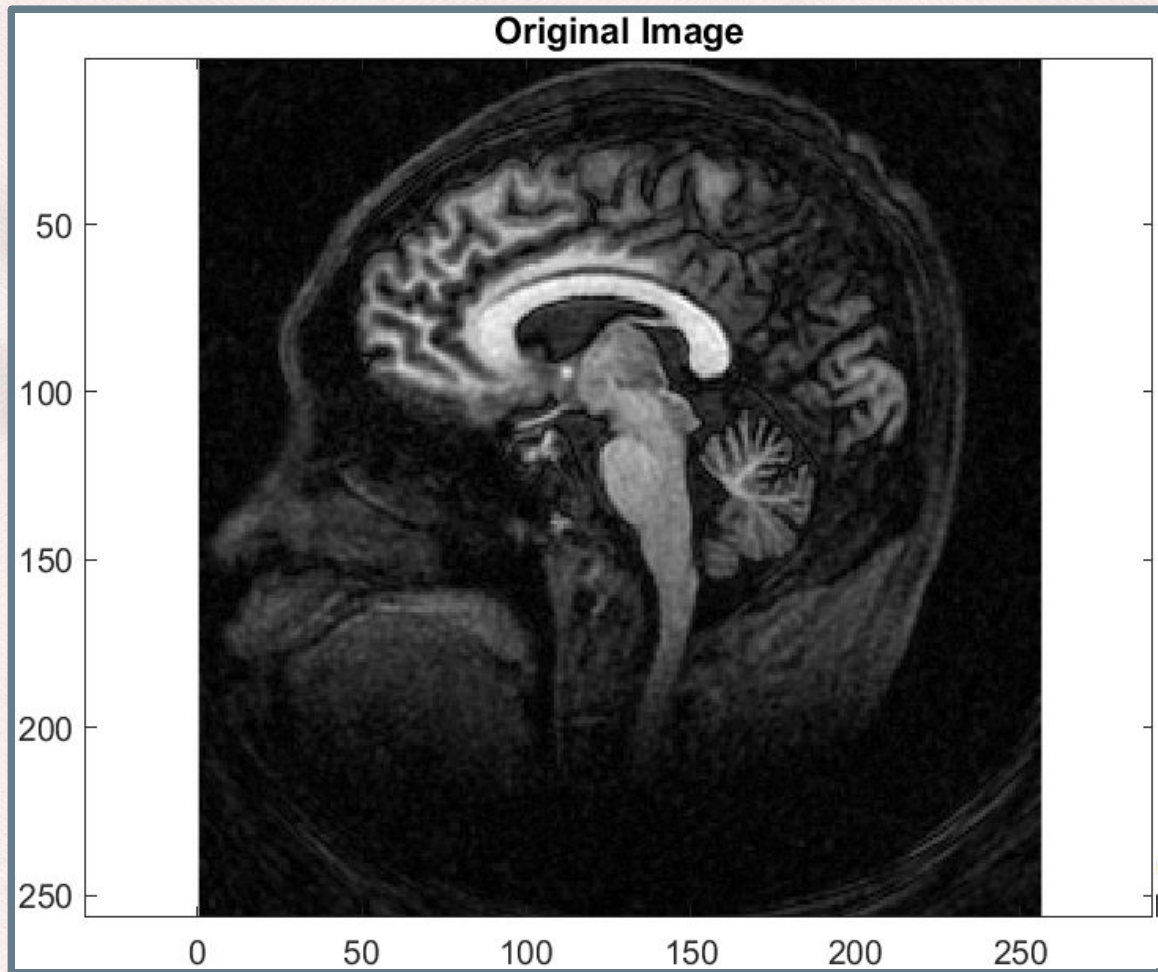


Reconstructed Image (idealized) (top 10% of k-space)



Reconstructed Image (Algor 2 Method) (top 10% of k-space)





$$RMSE = \sqrt{\frac{1}{n^2} \sum_i^n \sum_j^n ((x_i, y_j)_{\text{partial}} - (x_i, y_j)_{\text{original}})^2}$$

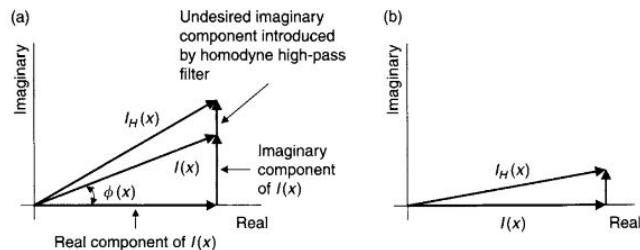


Homodyne Processing

What is Homodyne Processing?

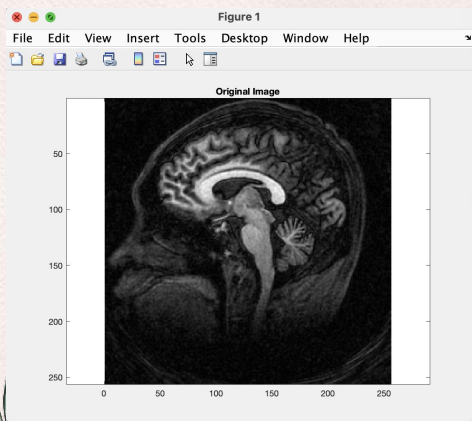
$$I(x) = \text{Re} \left[\int_{-k_0}^{k_0} S(k) e^{i2\pi kx} dk + 2 \int_{k_0}^{k_{\max}} S(k) e^{i2\pi kx} dk \right]$$

- ★ It uses a low spatial frequency map to correct for phase errors, exploiting the Hermitian Symmetry of K-Space Data simplified below:
- 1. Hermitian Conjugate Replacement:
 - a. Replace reconstructed image data with complex conjugate, using the fact that the complex conjugate is twice it's real part
 - b. Use a Homodyne Filter to simplify term, aka re-weighting vector. This allows for missing data to be zero filled and doubles weight of asymmetric data.
 - c. The Doubling makes it such that we have to consider only the real part since it generates unwanted imaginary component, using the fact that the complex conjugate is twice it's real part.
 - d. The abrupt transitions in the re-weighting vector cause ringing. A smoother function is used to reduce ringing.
- 2. Correction of Imaginary component: Some of the desired signal in the phase will be discarded using the above method, thus phase must be corrected using a low pass filter. [6]

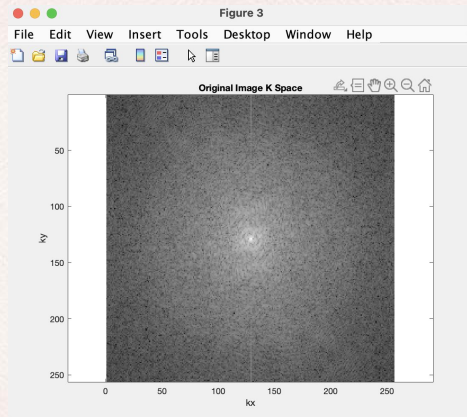


$$I(x) \approx \text{Re}[I_H(x) e^{-i\phi_L(x)}]$$

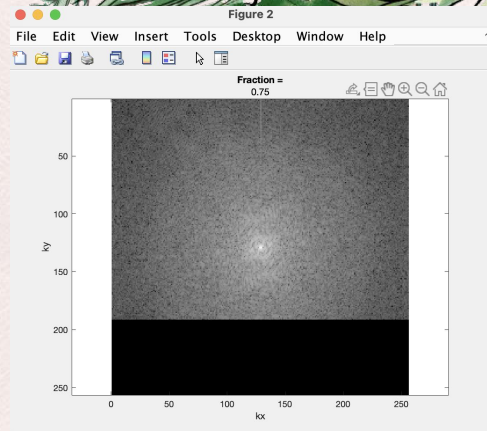
Our Application Method



Fourier transform



Use Partial K -Space



Multiply by the re-weighting vector

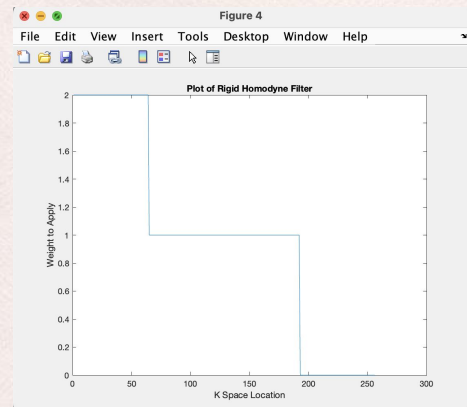
$b = \max(\text{real}(a), 0);$

Consider the Real Aspect of the output only

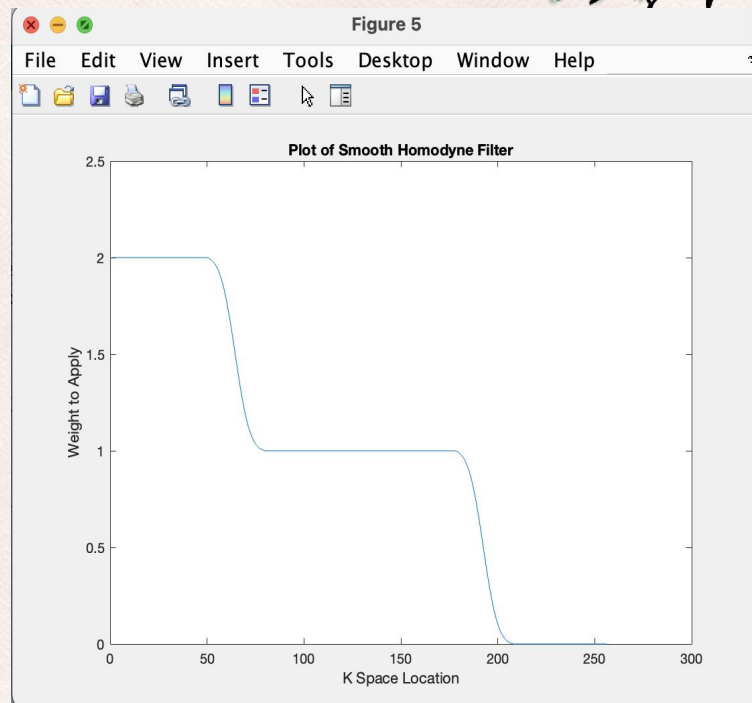
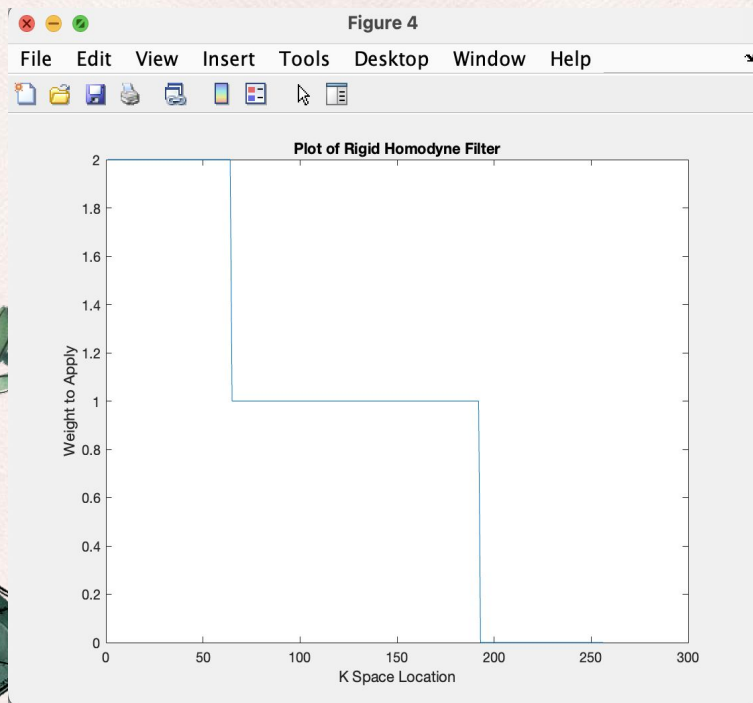
```
h = zeros(256);
h(64:192,:) = pM(64:192,:).*hann(129);
h = exp(1j.*angle(ifftdim(h,1:2)));
```

Inverse FT then Phase correction

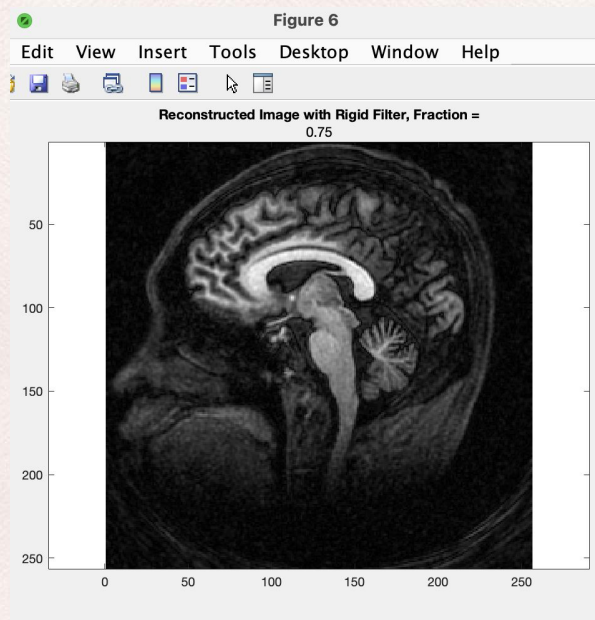
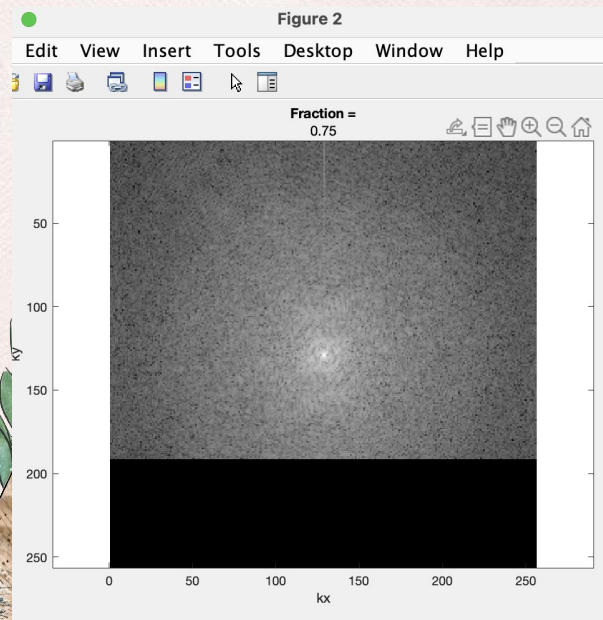
```
img_homodyne = real_nonneg(conj(h).* ifftdim(N,1:2));
```



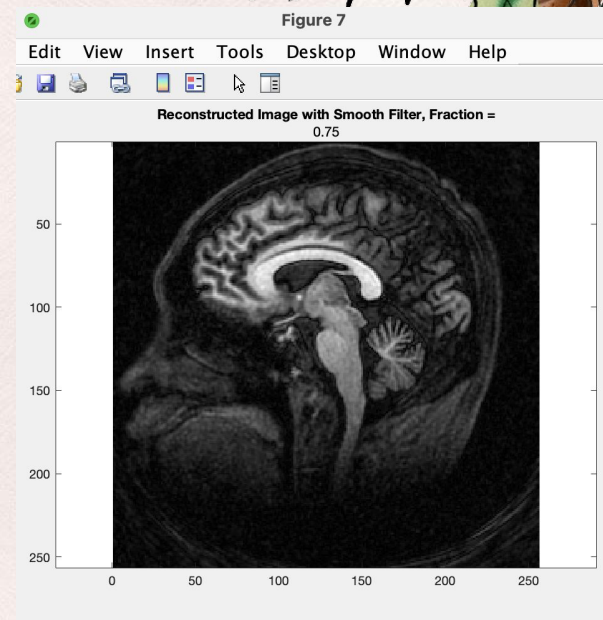
Re-Weighting Vectors Used



K-Space Fraction = 0.75

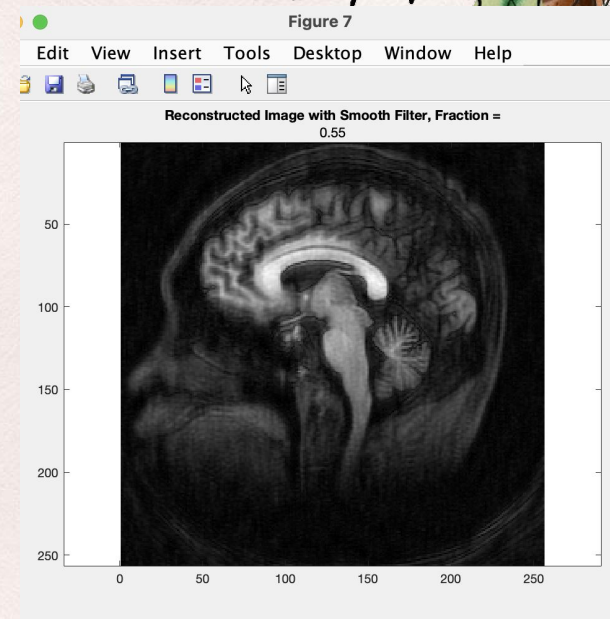
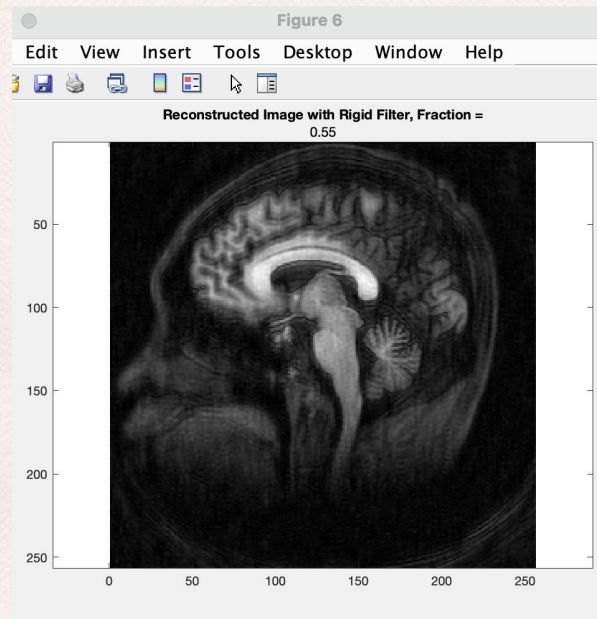
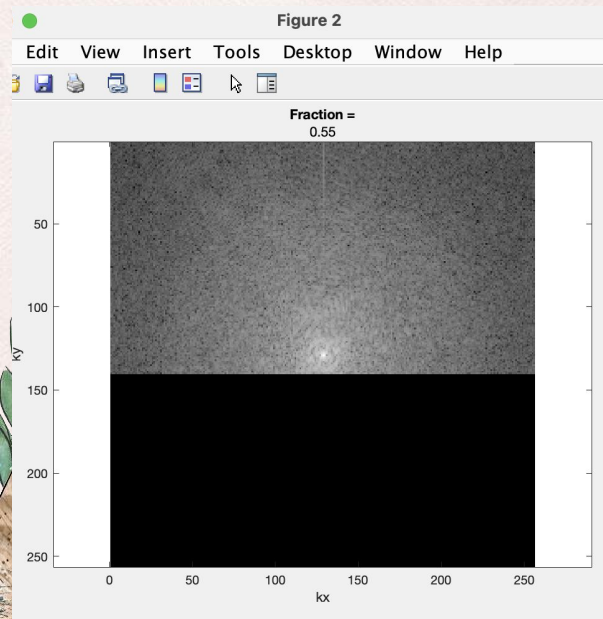


RMSE = 0.0135



RMSE = 0.0108

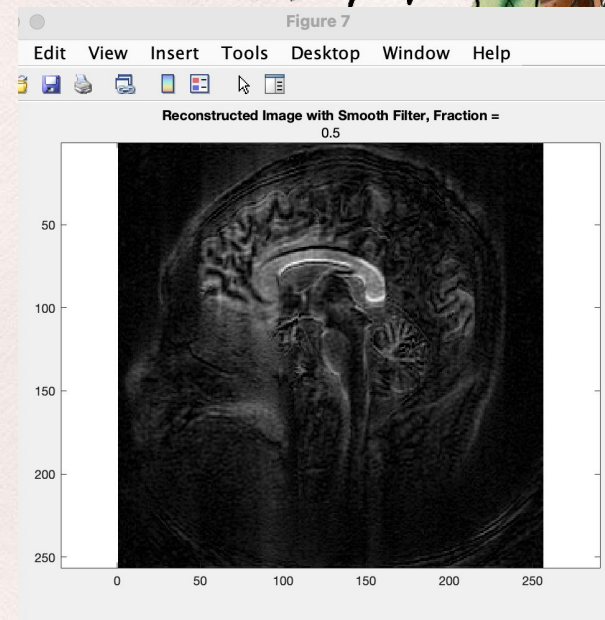
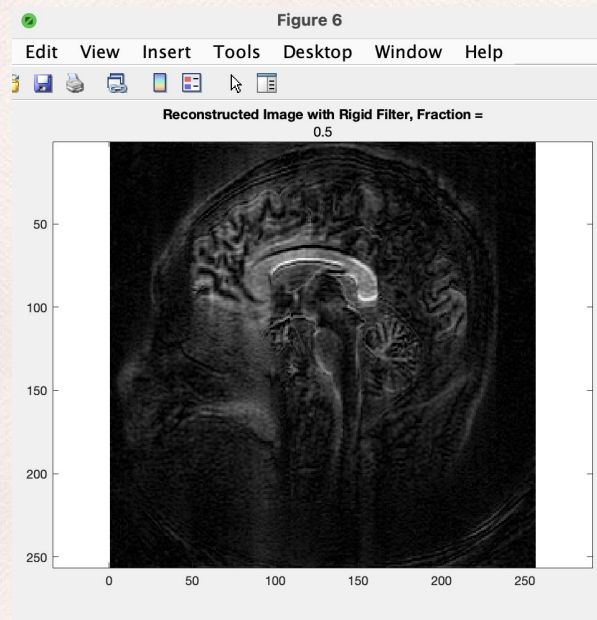
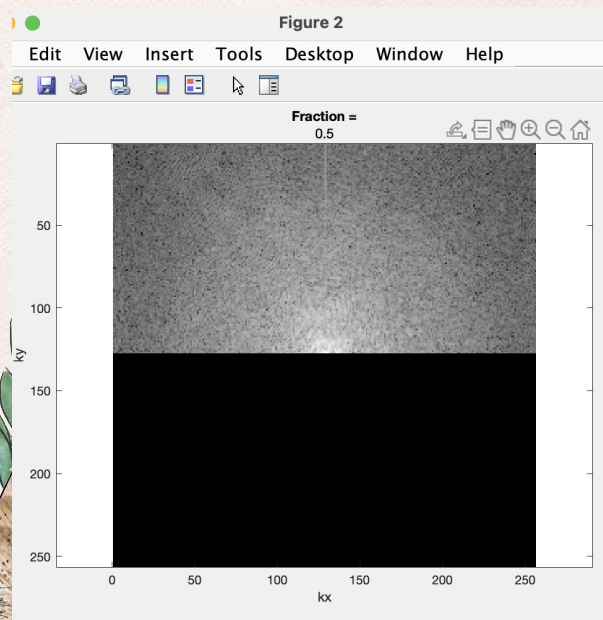
K-Space Fraction = 0.55



RMSE = 0.1439

RMSE = 0.1428

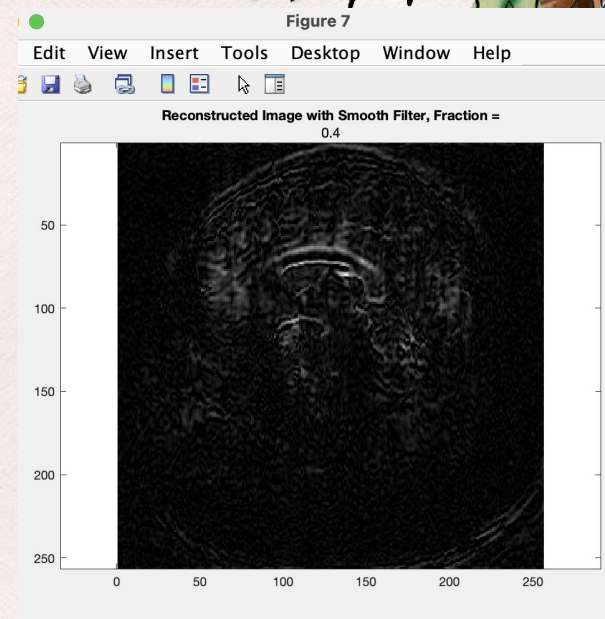
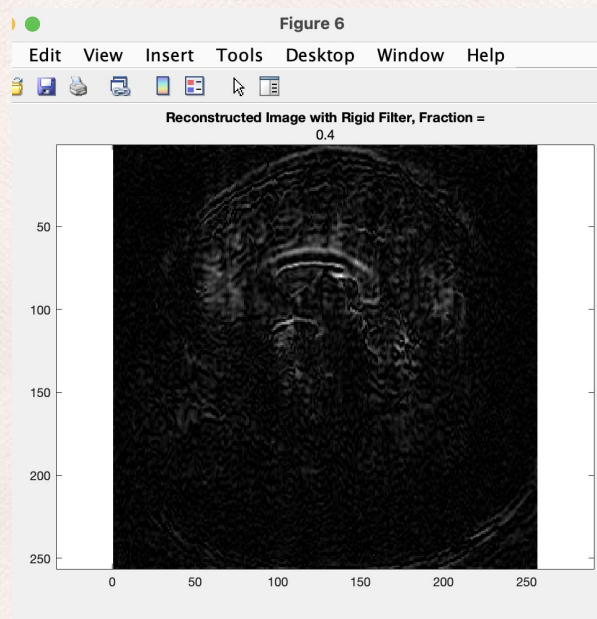
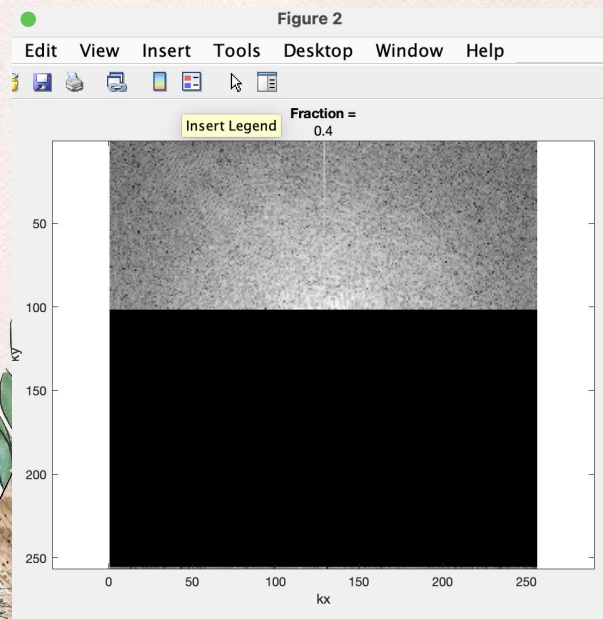
K-Space Fraction = 0.5



RMSE = 0.7408

RMSE = 0.7407

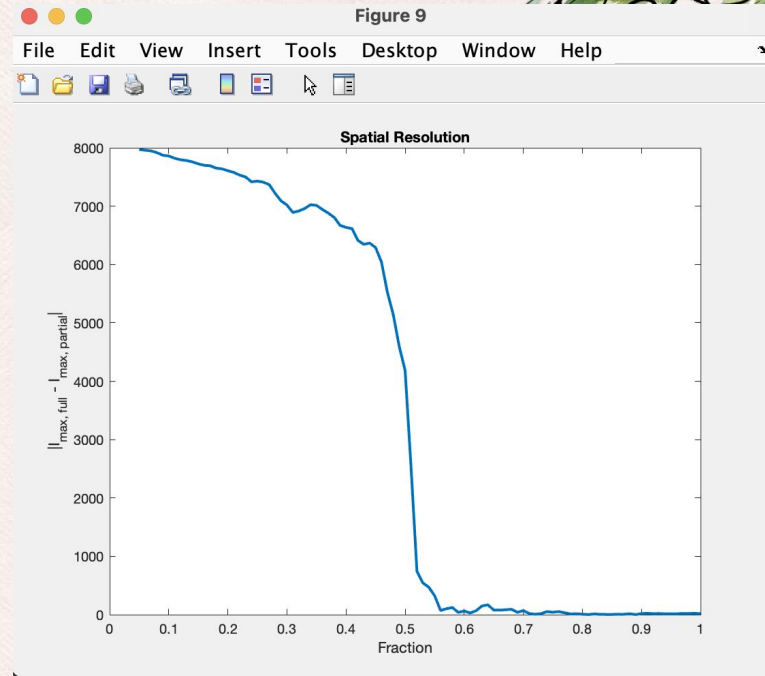
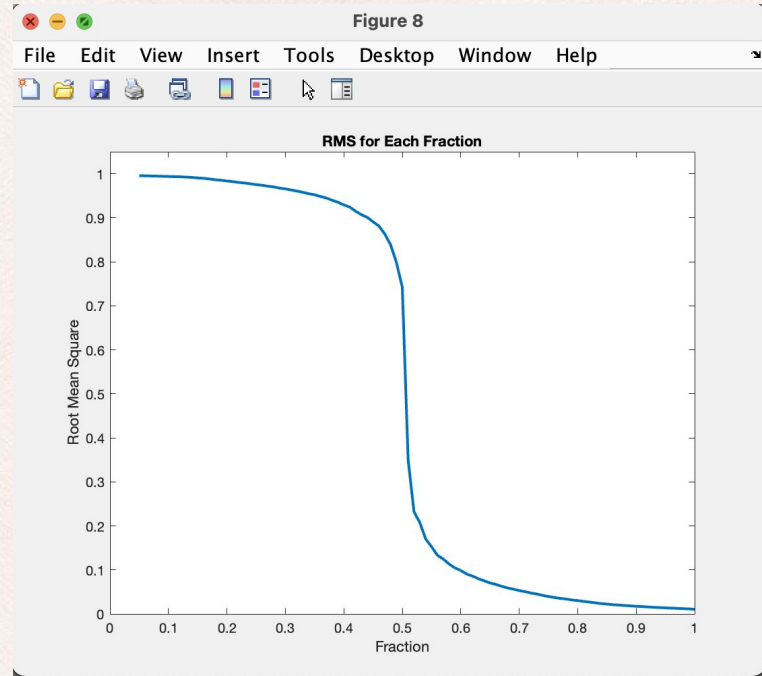
K-Space Fraction = 0.4



RMSE = 0.9211

RMSE = 0.9211

RMS and Resolution Plots for all fractions of K-Space





Conclusion and Future Work

Conclusions and Future Work

- Zero-filling maintains phase properties at high fractions, but gives rise to Gibbs ringing artifacts and relies on a high sample of data to produce quality images
- Conjugate synthesis does not preserve complex parts but is better in larger k-space since the real parts are symmetrical
- Homodyne changes phase properties to adjust for real components using high - and low - pass filters, still requires a significant amount of k-space data to have meaningful results
- For future prospects we can investigate the time advantage in greater detail of these methods
- We can also consider simulating partial fourier with parallel imaging models[10]

References

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2. Berger A. Magnetic resonance imaging. *BMJ*. 2002 Jan 5;324(7328):35. doi: 10.1136/bmj.324.7328.35. PMID: 11777806; PMCID: PMC1121941.
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5. *Partial Fourier*. Questions and Answers in MRI. (n.d.). Retrieved December 5, 2022, from <https://mriquestions.com/partial-fourier.html>
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8. 2-D inverse fast Fourier transform - MATLAB ifft2 [Internet]. [cited 2022 Dec 5]; Available from: <https://www.mathworks.com/help/matlab/ref/fft2.html>
9. Chiew Mark. Partial Fourier [Internet]. users.fmrib.ox.ac.uk/~mchiew/docs/PartialFourier. Available from: <https://users.fmrib.ox.ac.uk/~mchiew/docs/PartialFourier.html>
10. Ohliger, M.A., Larson, P.E., Bok, R.A., Shin, P., Hu, S., Tropp, J., Robb, F., Carvajal, L., Nelson, S.J., Kurhanewicz, J. and Vigneron, D.B. (2013), Combined parallel and partial fourier MR reconstruction for accelerated 8-channel hyperpolarized carbon-13 in vivo magnetic resonance Spectroscopic imaging (MRSI). *J. Magn. Reson. Imaging*, 38: 701-713. <https://doi.org/10.1002/jmri.23989>